

Assignment
Lesson: Basics of Probability Theory
Second Semester, SY 2025 - 2026

General Instructions:

1. A computer system requires a two-character login code. The code is made using the capital letters A , B , and C and the numbers 1, 2, and 3. Each login code consists of one letter followed by one number. Using the roster method, list all possible outcomes of the following events. [1 point each]
 - (a) The sample space (S)
 - (b) The event that the login code contains the number 1 (A)
 - (c) The event that the login code does not contain the number 1
 - (d) The event that the login code begins with the letter A (B)
 - (e) The event that the login code either contains the number 1 or begins with the letter A
 - (f) The event that the login code neither contains the number 1 nor the letter A
2. A cloud server schedules 4 different data queries: Query A , B , C , and D . The server executes them one at a time in a specific order.
 - (a) How many different execution orders are possible? [1 point]
 - (b) How many execution orders are possible if Query A and Query B must be executed consecutively? [2 point]
3. During data transmission, a system sends a packet containing the characters D, A, T, A, A . Because of network encoding, the characters may appear in different arrangements during packet reconstruction.
 - (a) How many distinct arrangements of the packet are possible? [1 point]
 - (b) How many distinct arrangements are possible if all the A 's must appear next to one another? [2 points]
4. A software engineering team is forming a task force for a project. There are 5 frontend developers and 4 backend developers available. In how many ways can a group of 5 developers be formed, consisting of 3 frontend developers and 2 backend developers? [2 points]
5. A cybersecurity team generates access tokens using 3 uppercase letters chosen from A, B, C , and D followed by 2 digits chosen from 1–5, in which characters cannot be repeated.
 - (a) How many access tokens are possible? [1 point]
 - (b) How many tokens start with A ? [1 point]
 - (c) How many tokens end with 5? [1 point]
 - (d) How many tokens both start with A and end with 5? [2 points]
6. From the previous item [2], define the following events:
 - Event A : The token starts with A
 - Event B : The token ends with 5

Determine the probability of the following:

- (a) $P(A)$ [1 point]
- (b) $P(A')$ [1 point]
- (c) $P(B)$ [1 point]
- (d) $P(A \cap B)$ [2 points]
- (e) $P(A \cup B)$ [2 points]

7. Interest centers around the uptime of a cloud server. Suppose the probability that the server remains operational for more than 6000 hours is 0.42, and the probability that it remains operational for more than 4000 hours is 0.65. In addition, given that the server operates for at most 6000 hours, the probability that it experiences a critical failure during its operation is 0.30.
- What is the probability that the server's uptime is less than or equal to 6000 hours? [1 point]
 - What is the probability that the server's uptime is between 4000 and 6000 hours? [2 points]
 - What is the probability that the server operates for at most 6000 hours and experiences a critical failure? [2 points]
8. In a computer network monitoring system, the probability that a server experiences high CPU usage is $\frac{2}{5}$, the probability that a server experiences high memory usage is $\frac{1}{3}$, and the probability that both occur simultaneously is $\frac{1}{5}$.
- Determine the following:
- The probability that the server will not experience high CPU usage [1 point]
 - The probability that the server will not experience high memory usage [1 point]
 - The probability that the server experiences at least one of the two issues [2 points]
 - The probability that a server experiences high CPU usage given that it experiences high memory usage [2 points]
 - The probability that the server experiences neither issue [3 points]
 - The probability that the server will not experience high memory usage given that it is experiencing high CPU usage [2 points]
 - The probability that the server will not experience high CPU usage given that it is experiencing high memory usage [2 points]
 - Determine whether high CPU usage and high memory usage are dependent or independent events [2 points]
9. A distributed system monitors two types of failures. The probability that a network timeout occurs is $\frac{1}{4}$, and the probability that a database failure occurs is $\frac{1}{3}$. If a network timeout is independent of a database failure, what is the probability that both failures occur? [2 points]
10. A cloud security system detects suspicious login activity using three monitoring modules: Module *A*, Module *B*, and Module *C*. Historical data show that:
- $\frac{1}{3}$ of login attempts are processed by Module *A*,
 - $\frac{1}{3}$ are processed by Module *B*, and
 - $\frac{1}{3}$ are processed by Module *C*.

The probabilities that each module flags a login attempt as suspicious are $\frac{1}{2}$, $\frac{1}{4}$, and $\frac{3}{4}$ for Modules *A*, *B*, and *C*, respectively.

- What is the probability that a randomly selected login attempt is flagged as suspicious? [3 points]
- What is the probability that a randomly selected login attempt is processed by Module *A* and flagged as suspicious? [2 points]
- If a randomly selected login attempt is flagged as suspicious, what is the probability that it was processed by Module *A*? [2 points]